DHARMIT SHAH

Ahmedabad, India — +91 99256 98090
GitHub ◇ Email ◇ LinkedIn ◇ Blog ◇ StackOverflow

SUMMARY

Infrastructure Software Engineer focusing on Kubernetes, writing k8s Operators in Go, and deep-diving into Linux internals and systems programming. Eager to leverage these to solve complex challenges and work on impactful projects.

SKILLS

Languages	Go (Golang), Python, Bash/Shell
Kubernetes & Cloud Native Ecosystem	Containers, Kubernetes Internals, Operators/CRDs, KubeVirt, Rancher, Harvester, Cluster API (CAPI), Tilt.dev, Helm, Admission Controllers / Webhooks
Systems & Infrastructure	Linux Internals (cgroups, namespaces, networking), Ignition, Butane, Libvirt / KVM, systemd
Tooling & CI/CD	Git, Bazel, GitHub Actions, Jenkins Pipelines, Ansible, Terraform

Upstream Highlight: Developed feature for Kubernetes core (`kubectl`) adding JSON output formatting to `api-resources`; devised an alternate implementation adopted and praised by maintainers.

EXPERIENCE *(16 YEARS)*

Senior Software Engineer, SUSE **Apr 2024 - Nov 2026**

- Elemental3 and LCM
  - Designed and implemented the self-upgrade reconciliation loop within the Kubernetes Operator that is used to manage lifecycle of immutable OS shipped as OCI image
  - Stack: Go, Kubernetes, Helm, Linux, Ignition, Butane
- Harvester
  - Eliminated indefinite upgrade hangs in Harvester by optimizing the upgrade workflow written using Wrangler framework, ensuring failure detection and feedback within 5 minutes.
  - Developed webhook to prevent user from removing "Witness" node taint & label from the "Witness" node:
  - Harvester pull requests
  - Stack: Go, Harvester, Wrangler framework
- Rancher
  - Implemented validating webhook to prevent accidental deletion of a critical system namespace, preventing cluster corruption and downstream deployment failures.
  - Added long pending `log` feature to fluentd Helm chart that improved UX
  - Diagnosed and resolved a metrics collection bug in Rancher, authoring the patch adopted across downstream releases
  - Documentation for Telegram receiver for Alerting
  - Full list of pull requests to Rancher can be found here
- ClusterAPI (CAPI)
  - Improved UX for RKE2 control-plane rolling upgrades by excluding machines marked for deletion; bug and fix
  - Fixed label propagation in Cluster API Provider RKE2 by ensuring custom labels defined on `RKE2ControlPlane` resources are correctly inherited by the managed Machine resources, enabling compatibility with ClusterClass topology and MachineHealthChecks; bug and fix.
  - Setup the never documented steps to attach debugger; updated the long outdated guide for developers new to the project; and a guide to deploy CAPI cluster to custom namespace.
  - Stack: Tilt, Go, Rancher, Helm

Senior Software Engineer, Red Hat **Oct 2015 - Mar 2024**

- KubeVirt — Developer *Dec 2022 - Mar 2024*
  - Delivered end-to-end ConfigMap support for hook sidecars, enabling more flexible configuration injection for developers & operators. Prior to this, only Go programs could be used to modify VM definition; this feature enabled developers & operators to use Python and Bash.
  - Designed enhancement to collect logs for all images and implemented it by limiting number of goroutines using worker pool pattern in `oc must-gather` to prevent system from getting overwhelmed
  - Full list of contributions to KubeVirt can be found here
  - Stack: Go, KubeVirt, Bazel
- odo — Developer and Team Lead (Nov 2020 - Dec 2021) *Apr 2019 - Dec 2022*
  - Concurrent log fetching to speed things up (12s to 1.5s); manually keep tab on running goroutines using `sync/atomic` internals
  - Lead the documentation effort: (website)
  - Lead the effort to educate technical support team by creating 12 videos; opened 18 bugs in the process, and received an award for the same
  - Stack: Go, spf13/cobra (CLI framework); goroutines, mutexes, `sync/atomic` (concurrency)
- CentOS Container Pipeline — Developer and Architect *Mar 2016 - Apr 2019*
  - Service written in Python and deployed on OpenShift using Ansible
  - Lead the architectural effort to move to OpenShift, thus moving the service into a single platform
  - Presented it at conferences, and architected its v2 based on OpenShift 3
  - Stack: Python; Ansible & OpenShift (deployment); Jenkins Pipelines (build orchestration), beanstalkd (message queue); Prometheus & Grafana (monitoring)

Earlier Roles — 2010-2015

- DevOps Engineer, Ishi Systems (Mesos, AWS, Terraform)
- Backend Engineer, Aubergine Systems (Django, AWS)
- Associate Software Engineer, Red Hat (Docker, Python)
- Associate Technical Support Engineer, Red Hat (Linux, Virtualization, OpenStack)
- Linux Administrator, Global SMSC (Firewall & `iptables`)
- IT Engineer, CMC Ltd.

EDUCATION

Bachelor of Engineering, Information Technology 2006 - 2010
Sardar Patel University, Anand

AWARDS & CERTIFICATIONS

Red Hat Developer Impact Award **Dec 2019**
Awarded for using the tool from a user's perspective, creating learning material for Technical Support team, and opening numerous bugs in the process

Certified Kubernetes Administrator **Nov 2020**

Certified Kubernetes Application Developer **Nov 2020**

COMMUNITY & MENTORING

- Lead of Kubernetes, Go and Docker meetup chapters in Ahmedabad
- Co-organizer for KCD Gujarat 2026
- Mentored undergraduate students in Linux & Python; mentees now work as engineers across Red Hat, SUSE, Oracle, and Intel (references available on request).

CONFERENCE TALKS

- DevConf.CZ 2018 and 2019 at Brno, Czech Republic
- CentOS Dojo 2019 at Brussels, Belgium